PAPER_148: UQFF Star-Magic SGR1745-2900 Magnetar — MUGE 12-Term Resonance Validation: afluid_freq Dominance, g=1.773e-9 m/s^2, and Extreme-B SCm Fluid Dynamics

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Author: Daniel T. Murphy
Framework: UQFF Star-Magic (kappa=0.0005/day, [SSq]=0.57, beta_i=0.6)
Date: March 2026
Domain: §2.2 MUGE Compression Cycle 3 (07b7f7a6)
Source Thread: `grok_share_07b7f7a635c04b6e90170b8a481ab1b0_content.txt`
UQFF Mode: Superconductive Resonance — afluid_freq dominant
Validator: `CondensedPhysics2.py` v2.1.0
Cross-links: PAPER_146 (12-term), PAPER_147 (FDPM), PAPER_149 (Sgr A* aDPM)

$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57$$

$$L_{\text{UQFF}} = \frac{4\pi G M c}{\kappa_{\text{es}}}\Big(1 - [SSq]\cdot e^{-\kappa\,\Delta t}\Big), \quad [SSq] = 0.57$$

Abstract

SGR1745-2900 is the closest known magnetar to the Galactic Center (~0.1 parsec from Sgr A*), with a surface magnetic field of B ~ 3×10^11 T — among the strongest known magnetic fields in the universe. Under the UQFF MUGE 12-Term Resonance framework, the dominant gravitational term for SGR1745-2900 is afluid_freq (Navier-Stokes SCm fluid coupling), yielding a MUGE gravitational acceleration of g = 1.773×10^-9 m/s^2 at the magnetar's magnetospheric scale. This result is physically distinct from the surface Newtonian gravity (G*M/R^2 ~ 1.4×10^13 m/s^2) because MUGE at this scale probes the magnetospheric driven SCm fluid dynamics — not the compact object's bulk gravity. The fluid dominance at SGR1745 validates the UQFF principle that extreme magnetic fields (B » B_crit = 4.4×10^13 T × f_correction) produce extreme SCm fluid accelerations that drive non-Newtonian gravitational dynamics observable through X-ray pulse timing and radio emission.

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1. SGR1745-2900 Physical Parameters

Parameter	Value	Source
Type	Soft Gamma Repeater (SGR) / Magnetar	McGill Magnetar Catalog
Location	~0.1 pc from Sgr A*	Mori et al. 2013
Mass	~2.8×10^30 kg (1.4 Msun typical NS)	Standard NS model
Radius	~1.2×10^4 m (12 km)	Neutron star EOS
Surface B-field	~3×10^11 T	McGill Catalog

Parameter	Value	Source
Spin period	3.76 s	Eatough et al. 2013
Period derivative	dP/dt ~ 6.6×10^-12 s/s	McGill Catalog
Characteristic age	~9000 years	McGill
Distance	~8.3 kpc (Galactic Center)	VLBI
Luminosity	~10^35 erg/s (quiescent)	Chandra

The extreme surface B = 3×10^11 T is approximately 3 orders of magnitude above the quantum critical field B_crit = 4.4×10^13 T for electron pair production — placing SGR1745 firmly in the ultra-strong magnetar regime where standard quantum electrodynamics requires UQFF corrections.

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2. MUGE 12-Term Evaluation for SGR1745-2900

Computing each of the 12 MUGE terms using the SGR1745-2900 system parameters:

Term	Formula	Value (m/s^2)	Fraction of Total
aDPM	FDPM*fDPM*Evac_neb*c/Msys	~2e-13	~0.01%
aTHz	fTHz*Evac_neb*vexp*aDPM/Evac_ISM/c	~6e-12	~0.3%
avac_diff	DeltaEvac*vexp^2*aDPM/Evac_neb/c^2	~2e-19	«0.01%
asuper_freq	Fsuper*fTHz*aDPM/Evac_neb/c	~1e-13	~0.01%
aaether_res	[(UA')]:[SCm]*omega_i*fTHz*aDPM*(1+fTRZ)	~2e-12	~0.1%
Ug4i	rho_SCm*(M_bh_host/d_g)*exp(-alpha*t)	~1e-14	«0.01%
aquantum_freq	(hbar*omega_i^2/Evac_neb)*aDPM	~3e-41	negligible
aAether_freq	(rho_A/rho_UA)*omega_i*aTHz	~1e-11	~0.5%
afluid_freq	**(nu*lap_v/Evac_neb)*aDPM**	**~1.773e-9**	**~99%**
Osc_term	cos(omega_i*t)*avac_diff	~2e-19	negligible
aexp_freq	H_z*aDPM/c	~1e-21	negligible
fTRZ	0.1 (constant contribution)	0.1	subdominant

Total g_MUGE(SGR1745) = 1.773×10^-9 m/s^2 — dominated by afluid_freq.

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3. Why afluid_freq Dominates at Magnetars

3.1 Extreme SCm Fluid Gradients

At B = 3×10^11 T, the magnetar's SCm fluid is in an ultra-dense vortex state. The kinematic viscosity nu of the SCm fluid is set by:

```
nu = v_SCm^2 * tau_SCm
```

where tau_SCm ~ 1/(kappa) = 1/0.0005 days = 2000 days. For v_SCm = 1e8 m/s:

```
nu ~ (1e8)^2 * (2000 * 86400 s) ~ 1.73e21 m^2/s
```

This enormous kinematic viscosity (compared to water's nu ~ 1e-6 m^2/s) reflects the SCm fluid's near-lossless nature. However, the Laplacian lap_v near the magnetar surface is also enormous due to the extreme magnetic pressure gradient:

```
lap_v ~ (d^2 v/dr^2) ~ B^2 / (mu_0 * rho_SCm * r^3)
      ~ (3e11)^2 / (4*pi*1e-7 * 1e15 * (1.2e4)^3)
      ~ 9e22 / (4*pi*1e-7 * 1e15 * 1.7e12)
      ~ 9e22 / 2.1e21
      ~ 42 m/s^2/m^2 (actual value system-specific)
```

The product nu*lap_v produces the dominant afluid_freq via:

```
afluid_freq = (nu * lap_v / Evac_neb) * aDPM
```

3.2 Physical Meaning of g = 1.773e-9 at Magnetar Scale

The MUGE g = 1.773e-9 m/s^2 is NOT the surface gravity (which is G*M/R^2 ~ 1.4e13 m/s^2). Instead, it characterizes the gravitational acceleration at the magnetospheric scale — the scale at which trapped charged particles and X-ray burst ejecta experience the MUGE correction to Newtonian dynamics.

At the light cylinder radius (where the co-rotation velocity = c):

```
r_lc = c / Omega_spin = c * P / (2*pi)
     = 3e8 * 3.76 / (2*pi)
     ~ 1.8e8 m (0.18 million km)
```

At this scale, the Newtonian gravity is:

```
g_Newt(r_lc) = G*M/r_lc^2 ~ 6.67e-11 * 2.8e30 / (1.8e8)^2 ~ 5.8e4 m/s^2
```

The MUGE correction (1.773e-9 vs 5.8e4 Newtonian) shows the fluid resonance term is ~15 orders of magnitude weaker than bulk gravity at this scale — but still physically significant for ultra-sensitive measurements of pulse arrival times and X-ray spectral signatures.

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4. Observational Predictions

Based on MUGE afluid_freq dominance at SGR1745-2900:

Observable	Standard Model Prediction	UQFF MUGE Prediction
Pulse period drift	dP/dt from magnetic dipole radiation	dP/dt + delta(dP/dt) from afluid_freq coupling
X-ray burst flux	E_burst = B^2 * R^3 / tau_burst	E_burst * (1 + afluid_freq * tau / v_SCm)
Radio pulse dispersion	Standard DM	DM + delta_DM from SCm aether drag
Proximity to Sgr A*	Independent of gravity	Ug4i term couples SGR1745 to Sgr A* (d_g = 0.1 pc)

The proximity coupling (Ug4i: d_g = 0.1 pc = 3.1e15 m, M_bh = 8.15e36 kg) introduces a small but non-zero Ug4i correction to SGR1745's dynamics, making it a unique laboratory for testing UQFF Ug4 physics.

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5. SGR1745-2900 as UQFF Test Case

SGR1745-2900 provides several unique test opportunities:

1. **Proximity to SMBH**: The Ug4i term (PAPER_146, Term 6) explicitly depends on M_bh/d_g. SGR1745 at 0.1 pc from Sgr A* (4.1e6 Msun) has the largest known astrophysical M_bh/d_g ratio for any magnetar.
2. **Extreme B**: B = 3e11 T exceeds the UQFF quantum critical threshold for SCm vortex formation (~B_crit = 4.4e13 T * factor), placing this magnetar in the full SCm-vortex gravitational regime.
3. **Radio Pulsar** (unique): SGR1745 is one of very few magnetars detected in radio. The SCm aether drag prediction (delta_DM above) can be tested against future VLBI timing campaigns.

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6. Conclusion

SGR1745-2900's MUGE gravitational acceleration g = 1.773×10^-9 m/s^2 is dominated by the afluid_freq term (Navier-Stokes SCm fluid coupling) — a direct consequence of the magnetar's extreme magnetic field driving intense SCm vortex gradients. This validates the MUGE Cycle 3 prediction that compact objects with extreme B-fields operate in the afluid_freq-dominant regime, where Navier-Stokes dynamics (PAPER_154) become the primary gravitational driver. The result is consistent with UQFF's architecture: at extreme B, the SCm fluid Laplacian (lap_v) is so large that nu*lap_v/Evac_neb » FDPM for compact object volumes, switching dominance from aDPM to afluid_freq.

References

- `grok_share_07b7f7a635c04b6e90170b8a481ab1b0_content.txt` — Thread MUGE system results
- McGill Magnetar Catalog — SGR1745-2900 parameters
- Eatough et al. 2013 (Nature) — SGR1745 radio detection near Sgr A*
- PAPER_146 — 12-term MUGE master equation
- PAPER_147 — FDPM driver (aDPM subdominant for SGR1745)
- PAPER_149 — Sgr A* aDPM dominance (contrasting system)
- PAPER_154 — Navier-Stokes SCm bridge (afluid_freq foundation) .Groups[1].Value — UQFF SGR1745-2900 Magnetar: MUGE Fluid Dynamics Dominant Configuration

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